4. The *Sliding Tiles* puzzle consists of a rectangular grid with marked square cells. A cell can be empty or hold a tile with a letter on it. Figure 1 shows an example.

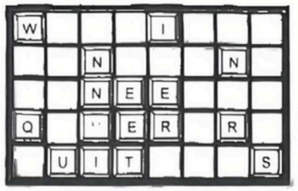

Figure 1

A player can slide a tile to a neighboring empty cell, up, down, left, or right. Let us call a configuration of tiles "packed" if no tile has a neighboring empty cell above or to the left of it. Figure 2 shows an example of a packed grid.

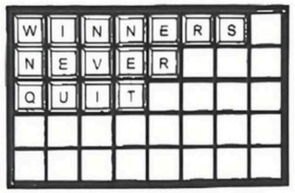

Figure 2

In a computer model of the game, the grid of cells is represented by a 2D array. A tile is represented by a one-letter literal `String` object; an empty cell holds a `null`.

(a) Write a static `boolean` method `isPacked` that returns `true` if a given configuration of tiles is "packed"; otherwise it returns `false`.

Complete the `isPacked` method below.

```
/** Returns true if the configuration of tiles represented
 *  by the 2D array tiles is packed, that is, any element
 *  that is not null does not have a null in the same
 *  column in the previous row or in the previous column
 *  in the same row; otherwise returns false.
 *  Precondition: tiles is not empty.
 */
public static boolean isPacked(String[][] tiles)
```

(b) To pack a given configuration of tiles, we can scan the whole grid row by row, starting at the top row, and each row from left to right. When we encounter a tile, we move the tile by sliding it up to the neighboring empty cell as far as possible, then to the left to the neighboring empty cell as far as possible. Write a private helper method `moveTile` that helps to implement this algorithm.

Complete the `moveTile` method below.

```
/** Relocates the tile at position (row, col) by first
 *  "sliding" it up several times to the previous row
 *  in the same column, as long as the cell there is empty;
 *  then sliding the tile to the left several times to
 *  the previous column in the same row, as long as the
 *  cell there is empty.
 *  Precondition: The 2D array tiles is not empty; the cell
 *                at location (row, col) holds a tile.
 */
private static void moveTile(String[][] tiles,
                                        int row, int col)
```